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Vellore Institute of Technology

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CHENNAI

22BAI1286

Final Assessment Test(FAT) - Apr/May 2025

Programme	**B.Tech.**	Semester	**Winter Semester 2024-25**
Course Code	**BHUM103L**	Faculty Name	**Prof. Shalina Susan Mathew**
Course Title	**Micro Economics**	Slot	**C1+TC1**
		Class Nbr	**CH2024250501365**
Time	**3 hours**	Max. Marks	**100**

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Describe traditional and modern definitions of economics.
CO2: Analyse supply and demand forces that determine equilibrium in a market economy.
CO3: Evaluate the factors affecting firm behaviour, such as production and costs.
CO4: Develop the skills to apply theories, models, and graphs to analyze the national and international cases.
CO5: Discuss the behaviour of market, industry and the performance of firms under different market structures.
CO6: Examine the market failures and the role of government in dealing with those failures.

Section - I

Answer any 7 Questions (7 × 10 Marks)

01. An article in a business magazine reported the discovery of a new processing technology that makes it economically feasible to turn natural gas into a liquid petroleum that yields petrol, diesel, or any other product derived from crude oil. This discovery represents 770 billion barrels of oil equivalent, 'enough to satisfy the world's thirst for oil for many years to come'. (Each 5 Marks)

(a) Using the demand and supply analysis, explain why this process will NOT cause a surplus of crude oil. If no surplus is created, what will be the impact of this process on the market for crude oil?
(b) In the unlikely event that no new oil deposits are ever discovered (including the one mentioned above), explain how the worldwide market for oil would have evolved.

[10] (CO2,4/K5)

02. Explain break-even analysis, with relevant diagram

[10] (CO3/K2)

03. Explain fixed cost, variable cost and marginal cost. With the help of a diagram, establish the relationship between average cost and marginal cost in short run production (You are not required to prepare a cost schedule). Explain the economic rationale for the shape of the marginal cost.

[10] (CO3/K2)

04. According to Lionel Robbins, "Economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses ". In a logical sequence, explain how the discipline of Economics addresses the ideas of scarcity, choice, trade-offs and opportunity cost. As an individual leading a student life, explain any two trade-offs and the corresponding opportunity cost that you face (Do not make your answer too personal, let your answers to the 'trade-offs you face' be general in nature).

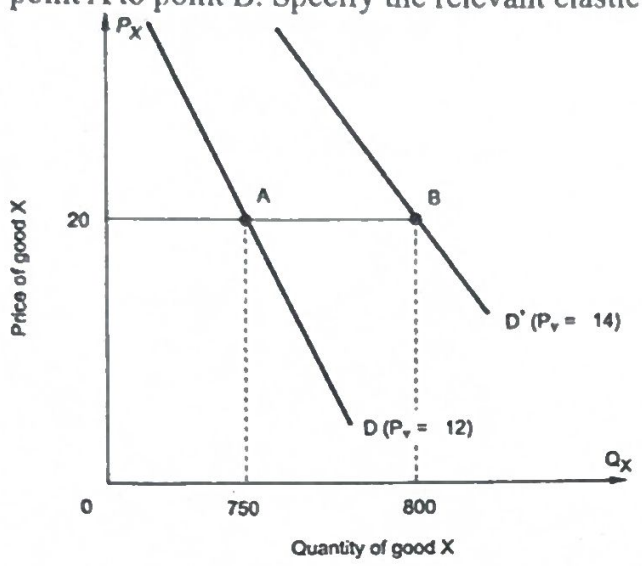
[10] (CO1/K4)

05. Binny reads in a business magazine that roadside dhabas on the Delhi-Agra Expressway are making a very large profit. Binny deides to open a dhaba on the sides of this Expressway. Unfortunately, several hundred other people also had read the article and had also rushed to open dhabas. While each dhaba is unique, Binny is making a loss. (Each 5 Marks)

(a) Describe the market structure where Binny operates. Show the market outcome facing Binny, with all the relevant details, in a graph.
(b) Explain how this market structure will adjust to long-run equilibrium. Provide relevant details in the graph.

[10] (CO4,5/K5)

06. (a) Use the concept of elasticity to explain each of the following observations. (5 Marks)
(i) During economic booms, when people are earning good wages, the number of new personal care businesses, such as gym and hair salons, are proportionately greater than the number of other new businesses such as grocery stores.
(ii) Some goods that were once considered luxuaries, such as mobile phone, are now considered necessities.
(b) Refer to the diagram. Calculate the relevant elasticity of demand for good X when it's demand shifts from point A to point B. Specify the relevant elasticity and it's value. (5 Marks)



[10] (CO2/K5)

07. Explain externality. With the help of relevant diagram, explain how taxation can be used as a tool for internalizing a negative externality.

[10] (CO6/K4)

08. Explain the concept of utility. With the help of a table, explain the concept of marginal utility. With the help of an example, explain how the consumption decisions that you make in your life reflect the idea of diminishing marginal utility.

[10] (CO2/K5)

Section - II
Answer any 2 Questions (2 × 15 Marks)

09. Define production function. With the help of relevant diagrams, explain how the producers arrive at the equilibrium of producing a given output at the minimum cost, in the long run.

[15] (CO3/K3)

10. With the help of relevant diagrams, explain the relationship between MR, AR, and price for a competitive firm and a monopoly firm. Explain the reasons for the observed relationship between these variables in the two types of market structures.

[15] (CO3/K1)

11. The following table shows the combinations of fixed and average variable cost depending upon the size of the plant chosen by the firm. (AVC is assumed to be constant for each plant size for the quantities considered in this example, i.e., it does not vary with the quantity of output produced).
(a) For each of these choices, calculate the ATC of producing 12,000 and 22,000 units of output. (4 Marks)
(b) Suppose that the firm, which has historically produced 12,000 units, experiences a sharp, permanent increase in demand that leads it to produce 22,000 units of output. (i) If the firm has to supply the required quantity in the short run, which plant size would it choose? (ii) How would the firm's decision change if it was making it's production decisions in the long run? Substantiate your answers based on relevant calculations.(8 Marks)
(c) Explain what the firm should do instead if it believes the change in the demand is temporary. Substantiate your answer based on relevant calculations. (3 Marks)

Plant size	Fixed Cost	Average Variable Cost
A	Rs. 8,000	Rs. 1.00
B	Rs. 12,000	Rs. 0.75

[15] (CO3,4/K5)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)